

Xianda Du

+1 226-752-4494 | a32du@uwaterloo.ca | [linkedin.com/in/xianda](https://www.linkedin.com/in/xianda) | www.allendu.me | github.com/XiandaDu

Education

University of Waterloo — CGPA until August 2025: 4.0 out of 4.0

Waterloo, ON

Bachelor of Applied Science in Computer Engineering, Option in Artificial Intelligence September 2022 – May 2027

- Undergraduate Research Assistant on FGSM and Gaussian Noise Analysis for ResNet-18 with Prof. En-Hui Yang.

Technical Skills

Languages: Python, JavaScript/TypeScript, HTML/CSS, Java, C/C++, SQL, Bash, R, MATLAB

Frameworks: React, Express, Vue, Tailwind, MaterialUI, Bootstrap, MantineUI, FastAPI, Spring Boot

Developer Tools: GCP, AWS, Cloudflare, Azure, Docker, K8s, Git, REST/GraphQL APIs, OAuth2, Scrapy, Nginx

Machine Learning: Transformers, RAG, CNN, LSTM, XGBoost, LightGBM, Scikit-learn, PyTorch, TF, Keras,

NumPy, Pandas, OpenCV, Matplotlib, (Hyperparameter Optimization)HPO, Jupyter Notebook

Database: Supabase, MySQL, MongoDB, Redis, Snowflake, Vector Database

Professional Experience

Research Assistant (NLP Benchmark)

Oct 2024 – Oct 2025

McGill University - RBC Borealis AI

Montreal, Quebec - Toronto, ON

- Co-first authored **WXImpactBench (ACL 2025)**, the first benchmark for LLMs' understanding of historical disruptive weather impacts, combining **NLP and meteorology**.
- Processed **53K+ OCR-scanned** articles with GPT-4o post-OCR correction, LDA topic modeling, and expert curation; annotated 1.7K samples across **six societal impact categories**.
- Designed **multi-label classification** and **ranking-based QA** tasks; benchmarked 12 LLMs (GPT, DeepSeek, LLaMA, Mistral, Qwen, Gemma) with metrics including F1, row-wise accuracy, Hit@1, and nDCG@5.
- Developed the **ranking-based QA** evaluation using **GPT-4o** for pseudo-question generation and a **sliding window re-ranking pipeline** implemented in **Python, HuggingFace Transformers, and NumPy/pandas**.
- Developing a **RAG study (ICLR-targeted)** creating a climate-domain benchmark to assess **retriever accuracy, climate adaptation, and generation quality** for improving LLM reliability in domain specific tasks.

WatAI RAG Project Technical Lead

April 2025 – December 2025

University of Waterloo, WatAI

Waterloo, ON

- Built a flexible **LangChain** layer that can switch between **Qwen, OpenAI, and Gemini**; supports **streaming** responses, **custom prompts**, per-user **API keys**, and simple guardrails (rate and token limits). | [Deployed App](#).
- Delivered a production **RAG** flow: ingest & chunk course PDFs/slides, embed and store in a **Supabase**, retrieve top passages (optional **hybrid dense+BM25**), **re-rank**, and answer with grounded **citations**.
- Added **agent** tools (retrieval, file parsing, syllabus/FAQ lookup) and a router that picks the best model by **speed, cost, and capability**; included timeouts, retries, and safe **fallbacks**.
- Set up **evaluation & monitoring**: retrieval metrics (**Recall@k, nDCG**), **LLM-as-judge** faithfulness checks, and run tracing (model, prompt version, retriever settings) to tune RAG with real chats.

Platform Developer

September 2024 – December 2024

University of Waterloo, ECE Department

Waterloo, ON

- Fine-tuned **Hugging Face** sentiment analysis models in Python using **Tensorflow/PyTorch** and analyzed results to improve media content classification, achieving **80%** accuracy.
- Used **pandas** to manage and label collected data, creating a pipeline to convert embeddings and data from a **PostgreSQL** database on **GKE** into a self-designed **neural network (NN)** to classify scraped articles.

Projects

Web 3.0 Quantitative Trading Challenge

Jun 2025 – Nov 2025

- Framed 15-min crypto returns as **ranking** and optimized **Weighted Spearman**; trained **TabNet, XGBoost, MLP**, and **LightGBM** via **Optuna**—finalized **Opus LightGBM v8**, scored **0.57** on the public leaderboard.
- Implemented **leak-safe, rolling-window** features (RSI divergence, distance-to-high/low, volatility) via vectorized **NumPy/pandas**; importance screening kept **80** features, improving generalization/training speed.
- Built **8-model LightGBM ensemble** (varying depth/regularization/seeds/losses) with **weighted blending**; applied adaptive clipping/winsorization, normalization, smoothing to stabilize ranks and beat single models.

Digit Recognition

Feb 2024 – April 2024

- A **digit recognition project** that uses **TensorFlow** to train and evaluate a handwritten digit recognition model.
- Showcased proficiency and understanding in deep learning. Employed **Convolutional Neural Network, softmax activation, Adam optimization, and cross-entropy loss** for accurate classification of handwritten digits.